Resolution Geometry: The Organization of the Ladder

Dependency Maps, Hand-overs, Walls, Gates and Anti-Ladders for L0–L126, with Views for Physicists, Geometers and Laboratories

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Companion to *Resolution Geometry* (print edition)

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Abstract

Resolution Geometry states its results as a ladder of 127 rungs, L0–L126, with backward-only dependencies. Read as a list, the ladder hides its own shape. A reader cannot see:

- which branches exist and where one branch hands a result to another;
- which no-go results act as walls, and what opens them;
- which hypotheses gate which results, and what a single measurement actually assumes.

This companion makes that shape explicit. It is generated mechanically from the ladder table of the print edition by a deterministic script, which first validates the table: contiguity, backward-only dependencies, acyclicity, a single root, reachability, declared hypotheses only, and nothing depending on an open problem.

The paper then provides eight views:

(1) a *metro map* with 16 branch lanes and logical depth on the horizontal axis;
(2) a *branch and hand-over map* (89 cross-branch edges);
(3) a *wall register* (27 no-go rungs and what opens each);
(4) a *gate register* with an assumption matrix;
(5) the *longest chains*;
(6) *anti-ladders*: reverse dependency cones drawn with the target at the top;
(7) *audience views*, with a physics, geometry and laboratory reading of each target;
(8) *measurement pipelines*: eleven concrete tasks, each with its full assumption budget.

The views expose four facts that the list form hides.

- The longest logical ladder is not physical distance (11 steps). It is the dynamics–chaos chain L0→L92 (16 steps); through the assembly umbrella the deepest rung is L123 (24).
- 96 rungs inherit H-sound and H-compl solely through the edge L9→L18.
- The lock dispatcher (L112–L115) inherits the transport hypotheses H-cov and H-resp through L111.
- A hypothesis-free core of 21 rungs exists.

These findings are reported as audit proposals for the next print. They are not applied here.

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1 How to use this companion

Reader	Start with	Then
Physicist	anti-ladders of L109, L105, L90, L80 (§8); assumption matrix (Fig. 3)	gate register (§6): which declarations carry SI, media, response, covariance
Geometer	unconditional core (Fig. 4); metro lanes B0, B2, B3, B8, B14	anti-ladder of L119; walls L65, L70, L117–L122; open L124, L126
Laboratory / engineer	pipelines (§10); anti-ladder of L113	wall register (what will fail and how); criticality hubs (Table 9)
Auditor	validation (Table 1); audit findings (§12)	full register (Appendix A)

Citation remains by rung ("RG, L90"). This companion adds no theorem. It reorganizes the statement of record, and where it disagrees with the print it reports an audit finding instead of changing a rung.

2 Data model

Rungs and local status. Each rung carries the status printed in the ladder. A rung may carry more than one class (for example "I / N"):

- F, forced;
- I, imported;
- H, conditioned on named hypotheses;
- Γ, conditional on the effective symmetry group;
- N, no-go;
- O, open.

Effective status (inherited budget). A rung's *assumption budget* is the union of the hypotheses, and of Γ, over the rung and all its ancestors. A rung whose local status is F but whose budget is non-empty is *forced given* those inherited hypotheses. The register (Appendix A) prints both columns; the difference is the main source of misreading.

Edges. An edge $u \to v$ means that v depends on u.

- *Logical* edges are the printed dependencies, including short ranges such as L54–L56.
- The *assembly umbrella* L1–L109 of L110 is kept separate (109 edges). It aggregates modules rather than using each one, so it would make every rung an ancestor of the closure layer.

Branches. The ladder's 16 section blocks are the branches B0–B15, each tagged with a primary audience. A *hand-over* is a logical edge between branches.

Walls and gates. A *wall* is a rung with class N. A *gate* is a rung with class H or Γ; a gate opens the region beyond it under a declaration.

Anti-ladder. The anti-ladder of a target t is its ancestor cone, transitively reduced, and drawn with t at the top and the root L0 at the bottom. It answers the question *"what must hold for t?"*, which is the reverse of the ladder's *"what follows?"*. Red edges are hand-overs. The *forward cone* of a rung is its set of descendants: what falls if the rung fails.

Check	Result	Meaning
O1	pass	rungs are exactly L0–L126, contiguous
O2	pass	every dependency points backward (the print's ordering claim holds)
O3	pass	the dependency graph is acyclic
O4, O5	pass	L0 is the single root, and every rung is reachable from it
O6, O7	pass	every status parses; every hypothesis is a declared name
O8	pass	nothing depends on an open rung (L124–L126 are leaves)
O9	pass	L0 is the only input

Table 1: Validation of the printed ladder table (script `rg_ladder_org.py`).

The graph has 127 rungs, 217 logical edges (179 after transitive reduction), 109 umbrella edges and 89 hand-overs. Counting every class a rung carries, the classes are: 49 F, 28 I, 27 H, 27 N, 8 Γ, 3 O and 1 input.

3 View 1: the metro map

Figure 1 draws every rung once.

- **Lanes** are branches, shaded by audience: blue for geometry, green for physics, orange for laboratory, grey for shared.
- **Horizontal position** is logical depth, the length of the longest chain from L0.
- **Marker shape and colour** give the local class.
- **Grey edges** stay within a branch. **Coloured arcs** are hand-overs, coloured by the exporting branch.
- **The dashed line** marks the assembly umbrella into L110.

The map shows the architecture at a glance. Three hubs (L2, L18, L19) feed almost every lane. The physical-distance lane B12 collects hand-overs from seven branches. The chaos lane B11 sits at the far right because its chain is the longest.

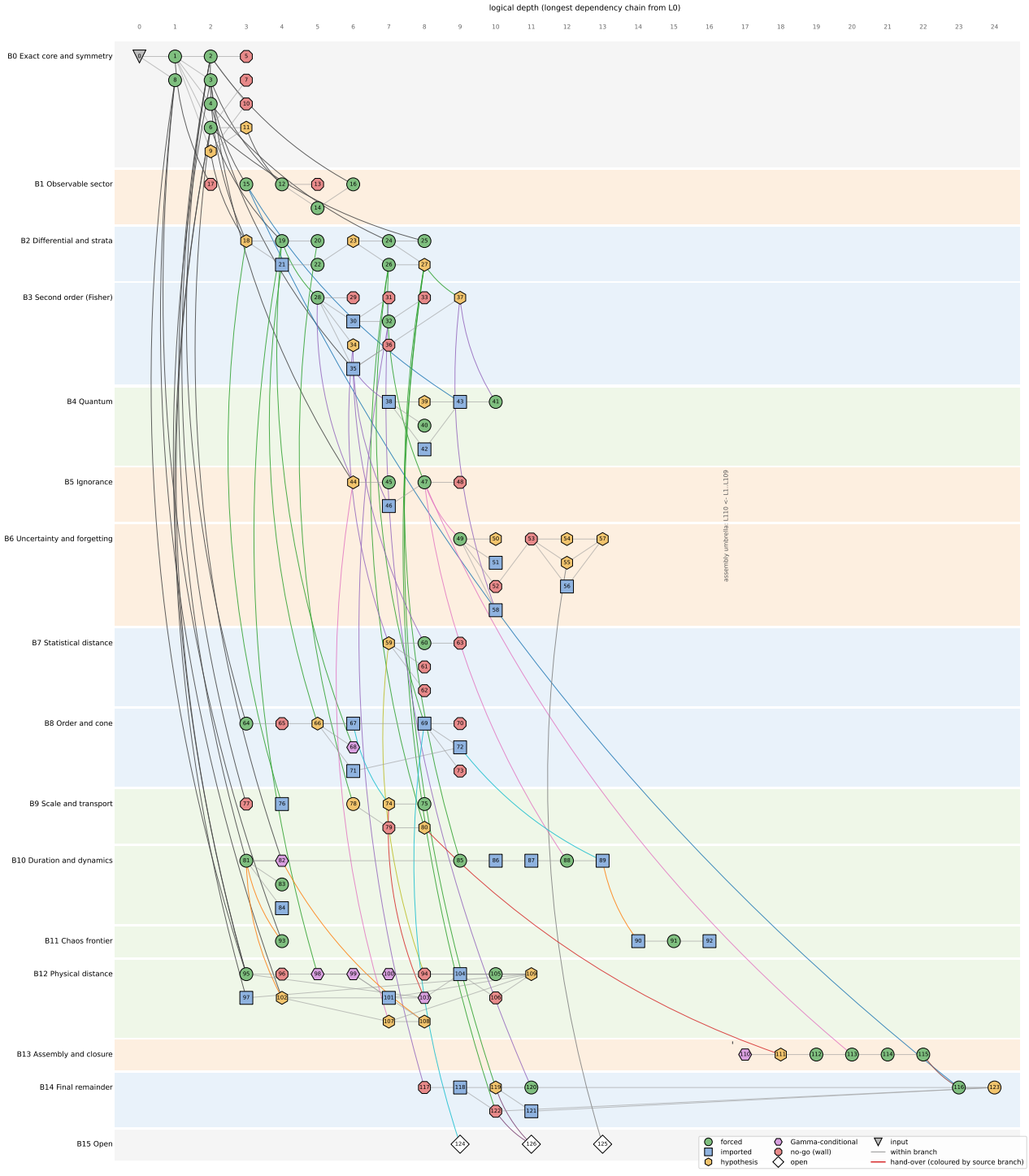


Figure 1: Metro map of L0–L126 (transitively reduced logical edges).

4 View 2: branches and hand-overs

Figure 2 collapses each branch to one node. An edge label gives the number of hand-over edges and the first exporting→importing rungs; the complete list is Table 3. Hand-overs are where one branch's guarantee becomes another branch's premise, so they are where misreadings concentrate.

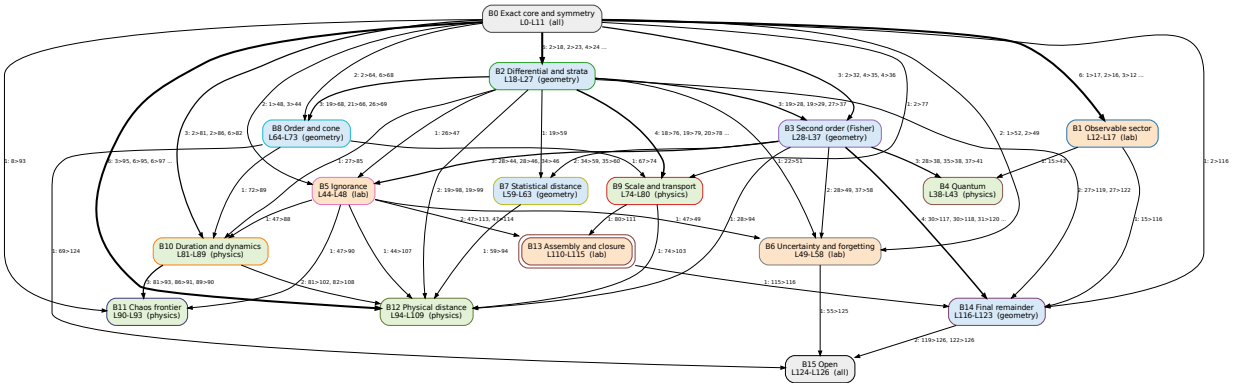


Figure 2: Branch quotient: 16 branches, 44 branch pairs, 89 hand-over edges. B13 receives in addition the assembly umbrella from every branch.

Hand-over	Edges	Exporting rung → importing rung
B0 → B1	6	L1 (kernel groupoid)→L17, L2 (universal factorization)→L16, L3 (joint kernel = intersection)→L12, L3 (joint kernel = intersection)→L14, L3 (joint kernel = intersection)→L15, L11 (coherent completeness)→L12
B0 → B10	3	L2 (universal factorization)→L81, L2 (universal factorization)→L86, L6 (Gau-Stab-Gamma sequence)→L82
B0 → B11	1	L8 (count rigidity)→L93
B0 → B12	6	L3 (joint kernel = intersection)→L95, L6 (Gau-Stab-Gamma sequence)→L95, L6 (Gau-Stab-Gamma sequence)→L97, L6 (Gau-Stab-Gamma sequence)→L98, L8 (count rigidity)→L95, L8 (count rigidity)→L102
B0 → B14	1	L2 (universal factorization)→L116
B0 → B2	6	L2 (universal factorization)→L18, L2 (universal factorization)→L23, L4 (garbling enlarges kernel)→L24, L6 (Gau-Stab-Gamma sequence)→L19, L6 (Gau-Stab-Gamma sequence)→L25, L9 (orbit-quotient comparison)→L18
B0 → B3	3	L2 (universal factorization)→L32, L4 (garbling enlarges kernel)→L35, L4 (garbling enlarges kernel)→L36
B0 → B5	2	L1 (kernel groupoid)→L48, L3 (joint kernel = intersection)→L44
B0 → B6	2	L1 (kernel groupoid)→L52, L2 (universal factorization)→L49
B0 → B8	2	L2 (universal factorization)→L64, L6 (Gau-Stab-Gamma sequence)→L68
B0 → B9	1	L2 (universal factorization)→L77
B1 → B14	1	L15 (coupled refines marginals)→L116
B1 → B4	1	L15 (coupled refines marginals)→L43
B10 → B11	3	L81 (clock groupoid)→L93, L86 (entropy from counts)→L91, L89 (chaos cone fields)→L90
B10 → B12	2	L81 (clock groupoid)→L102, L82 (affine clocks)→L108
B13 → B14	1	L115 (maximal closure)→L116
B14 → B15	2	L119 (global structure)→L126, L122 (wall monodromy)→L126
B2 → B10	1	L27 (stratified quotient category)→L85
B2 → B12	2	L19 (natural descent)→L98, L19 (natural descent)→L99
B2 → B14	2	L27 (stratified quotient category)→L119, L27 (stratified quotient category)→L122

Hand-over	Edges	Exporting rung → importing rung
B2 → B3	3	L19 (natural descent)→L28, L19 (natural descent)→L29, L27 (stratified quotient category)→L37
B2 → B5	1	L26 (tangent USC, tame strata)→L47
B2 → B6	1	L22 (one-sided wall directions)→L51
B2 → B7	1	L19 (natural descent)→L59
B2 → B8	3	L19 (natural descent)→L68, L21 (slices and tangent cones)→L66, L26 (tangent USC, tame strata)→L69
B2 → B9	4	L18 (tangent sequence)→L76, L19 (natural descent)→L79, L20 (automatic descent)→L78, L26 (tangent USC, tame strata)→L80
B3 → B12	1	L28 (Fisher natural)→L94
B3 → B14	4	L30 ((R,N,Sigma) reconstruction)→L117, L30 ((R,N,Sigma) reconstruction)→L118, L31 (Fisher does not fix covariance)→L120, L32 (quotient dim p+q+pq)→L117
B3 → B4	3	L28 (Fisher natural)→L38, L35 (data processing)→L38, L37 (Hellinger specialization)→L41
B3 → B5	3	L28 (Fisher natural)→L44, L28 (Fisher natural)→L46, L34 (efficient metric)→L46
B3 → B6	2	L28 (Fisher natural)→L49, L37 (Hellinger specialization)→L58
B3 → B7	2	L34 (efficient metric)→L59, L35 (data processing)→L60
B5 → B10	1	L47 (rank walls not certifiable)→L88
B5 → B11	1	L47 (rank walls not certifiable)→L90
B5 → B12	1	L44 (persistent null)→L107
B5 → B13	2	L47 (rank walls not certifiable)→L113, L47 (rank walls not certifiable)→L114
B5 → B6	1	L47 (rank walls not certifiable)→L49
B6 → B15	1	L55 (contraction gate)→L125
B7 → B12	1	L59 (RG distance)→L94
B8 → B10	1	L72 (hyperbolic lock)→L89
B8 → B15	1	L69 (tame reduction)→L124
B8 → B9	1	L67 (conformal ray)→L74
B9 → B12	1	L74 (scale cocycle)→L103
B9 → B13	1	L80 (canonical connection)→L111

Table 3: Every hand-over edge, listed by branch pair.

5 View 3: walls

A wall is a no-go rung: a proved impossibility with a witness. Table 5 lists each wall, its direct successors, and *what opens it*, which is the declaration, hypothesis or anchor that lets the ladder continue. The walls fall into three kinds:

- *terminal walls*, with no successor, which are permanent boundaries (for example L5, L13, L36, L48);
- *gated walls*, opened by a named hypothesis (for example L53 by H-adm, L65 by H-top, L79 by H-cov);
- *anchored walls*, opened only by counted channels and SI anchors (L94, L96, L63).

Wall	What it denies	Br.	Direct successors (status)	Opened by
L5	exact layer fixes nothing	B0	*terminal*	declared higher structure: H-reg (L18), H-mono (L29), H-top (L66), Γ, H-SI
L7	no splitting	B0	*terminal*	presentation-relative Γ (L6)
L10	rank, separation, completeness distinct	B0	*terminal*	separate certificate per property (L11, L12)

Wall	What it denies	Br.	Direct successors (status)	Opened by
L13	no task-free selector	B1	*terminal*	a declared task (L12 then selects exactly)
L17	tolerance not a groupoid	B1	*terminal*	confidence sets (L16); window walls (L114)
L29	Fisher selected (Cencov)	B3	*terminal*	H-mono selects Fisher (Čencov)
L31	Fisher does not fix covariance	B3	L120 (F)	second-order contract (L32, L117); global $[K_N]$ (L120)
L33	shadow is not the law	B3	*terminal*	protocols on the full law (L15, L43)
L36	kernel order not Blackwell	B3	*terminal*	none inside the ladder (decision-theoretic order)
L48	law equality not certifiable	B5	*terminal*	confidence correspondences (L16, L47)
L52	TV rigidity	B6	L53 (N)	a declared class (L53 to L57)
L53	no uniform forgetting	B6	L54 (H: adm); L55 (H: adm); L56 (I / N)	H-adm: BV lock (L54), contraction gate (L55)
L56	samples refute, operators lock	B6	L57 (H: adm)	operator certificates under H-adm (L57)
L61	no angle in rank one	B7	*terminal*	rank-two strata; anchors
L62	divergences not rulers	B7	*terminal*	infinitesimal metric (L59); anchors (L102)
L63	experiment-relative distance	B7	*terminal*	counted channel and SI anchor (L95, L102, L109)
L65	projective typing	B8	L66 (H: top)	H-top: realization branches (L66)
L70	no finite classification	B8	*terminal*	tame reduction (L69); enumeration open (L124)
L73	wild class	B8	*terminal*	window certificates only
L77	entropy selects no connection	B9	*terminal*	declared covariance (L80)
L79	no law-natural connection	B9	L80 (H: cov; resp)	H-cov: canonical connection (L80)
L94	no internal unit	B12	L109 (H: reg, nuis, SI, med, val; Γ)	counted channel (L95) and H-SI (L102)
L96	count loss	B12	L109 (H: reg, nuis, SI, med, val; Γ)	loss-calibrated channel (H-val in L109)
L106	non-stationary no-go	B12	L109 (H: reg, nuis, SI, med, val; Γ)	stationary slices only (L104); otherwise open
L117	coupling complement-relative	B14	L118 (I / N); L123 (F (H: split, type))	H-split: declared complement
L118	coupling stabilizer	B14	L119 (H: split, type); L122 (N / F); L123 (F (H: split, type))	H-type: constant multiplicity (L119)
L122	wall monodromy	B14	L123 (F (H: split, type)); L126 (O)	open across walls (L126)

Table 5: Wall register.

6 View 4: gates and the assumption budget

Table 6 lists each hypothesis, the rungs that state it, and the size of the logical region it gates. Figure 3 gives, for 24 audience targets, every hypothesis the target needs: red where the target itself states it, amber where it is inherited.

Hypothesis	Gated rungs	Downstream
H-compl	L9	102
H-sound	L9	102
H-reg	L18, L59, L109	93
H-unif	L23, L27, L37	43
H-nuis	L34, L59, L109	29
H-top	L66	19
Γ	L68, L82, L98, L99, L100, L103, L109, L110	12
H-resp	L80	7
H-SI	L102, L109	6
H-cov	L80, L111	6
H-int	L74	6
H-coh	L11	4
H-sens	L44	4
H-split	L119, L123	3
H-type	L119, L123	3
H-hell	L37	2
H-adm	L54, L55, L57	1
H-band	L121	1
H-med	L107, L109	1
H-task	L39	1
H-exch	L50	0
H-mono	L29	0
H-stat	L108	0
H-val	L109	0

Table 6: Gate register, ordered by the size of the gated region.

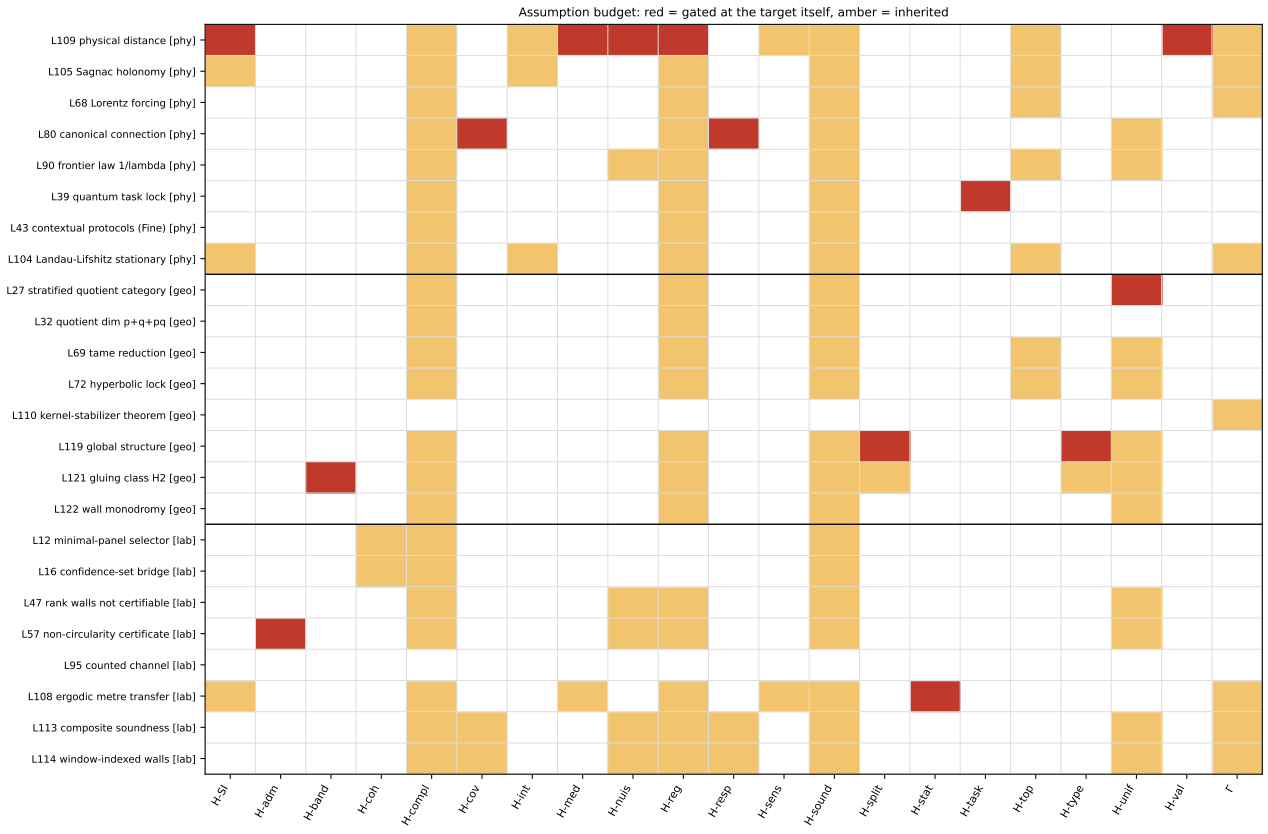


Figure 3: Assumption matrix for physics, geometry and laboratory targets.

The unconditional core. 21 rungs have an empty budget: no hypothesis and no Γ anywhere below them (Figure 4). They are L0–L8, L15, L17, L64, L65, L77, L81, L83, L84, L93 and L95–L97. This is what RG asserts with no declaration beyond the observation system itself:

- the kernel and quotient;
- the symmetry sequence;
- coupling and tolerance;
- reachability;
- the clock groupoid, path independence and distortion classes;
- the counted channel and the ratio scale.



Figure 4: The unconditional core (empty assumption budget).

7 View 5: longest chains

Target	Length	Longest logical dependency chain
L109	11	L0→L1→L2→L18→L21→L66→L67→L74→L103→L104→L105→L109
L113	11	L0→L1→L2→L18→L21→L22→L23→L26→L80→L111→L112→L113
L119	10	L0→L1→L2→L18→L19→L28→L30→L32→L117→L118→L119
L90	14	L0→L1→L2→L18→L21→L22→L23→L24→L27→L85→L86→L87→L88→L89→L90
L80	8	L0→L1→L2→L18→L21→L22→L23→L26→L80
L39	8	L0→L1→L2→L18→L19→L28→L35→L38→L39
L68	6	L0→L1→L2→L18→L21→L66→L68
L126	11	L0→L1→L2→L18→L19→L28→L30→L32→L117→L118→L119→L126
global	16	L0→L1→L2→L18→L21→L22→L23→L24→L27→L85→L86→L87→L88→L89→L90→ L91→L92

Table 8: Longest logical dependency chains.

Correction to a common reading. The longest ladder is the dynamics–chaos chain: 16 steps, through the strata (L18–L27) and the entropy and Pesin locks. It is not physical distance, which needs only 11 steps but has the widest cone outside the closure layer (34 ancestors from nine branches). Second

order reaches the remainder in 10 steps (L119). Through the assembly umbrella the deepest rung is L123, at 24 steps.

8 View 6: anti-ladders

Each figure answers *"what must hold for this target?"* The target is at the top (double border) and L0 at the bottom; red edges are hand-overs. Bracketed labels name hypotheses stated at that rung.

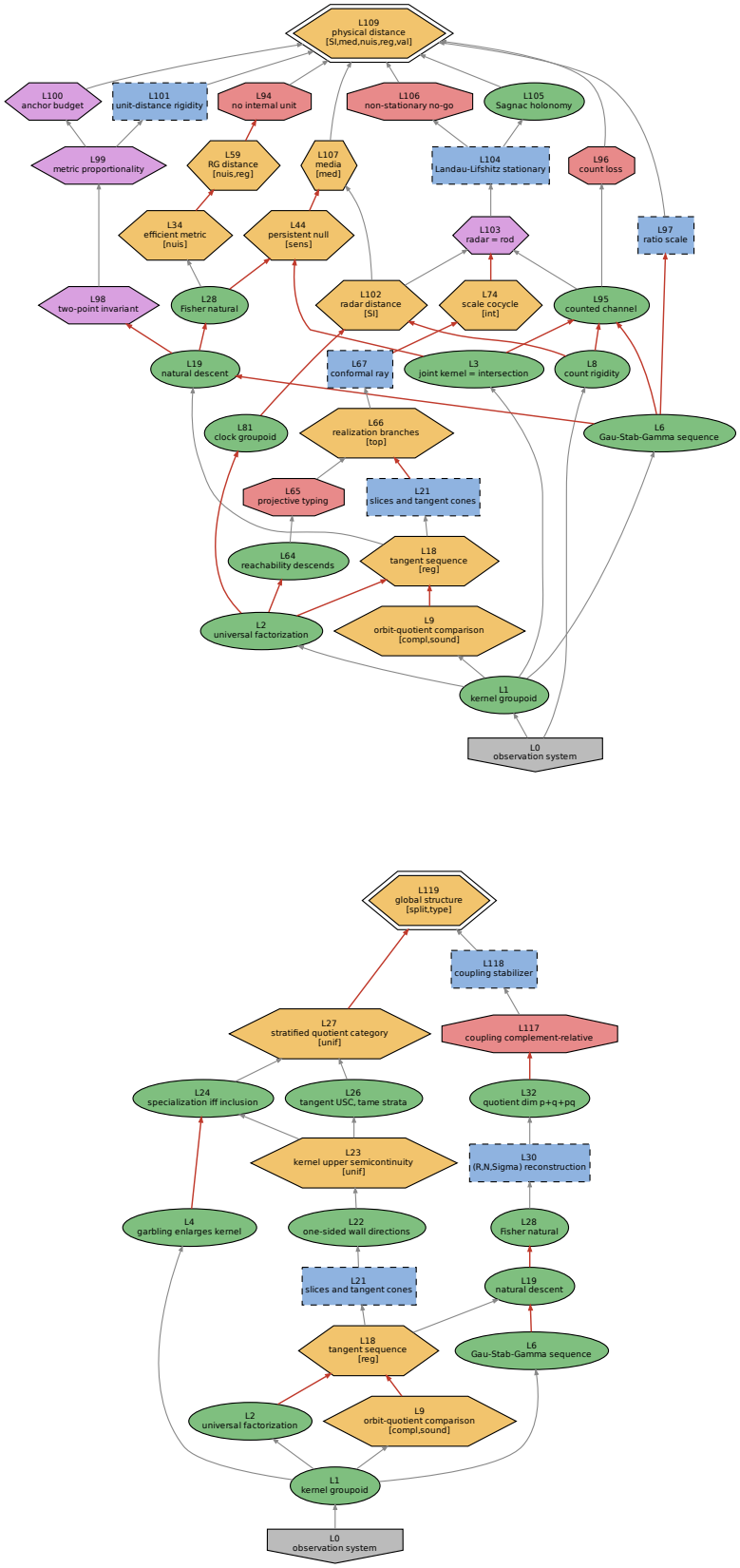


Figure 5: Anti-ladders of L109, physical distance (physics), and L119, the global structure of the second-order remainder (geometry).

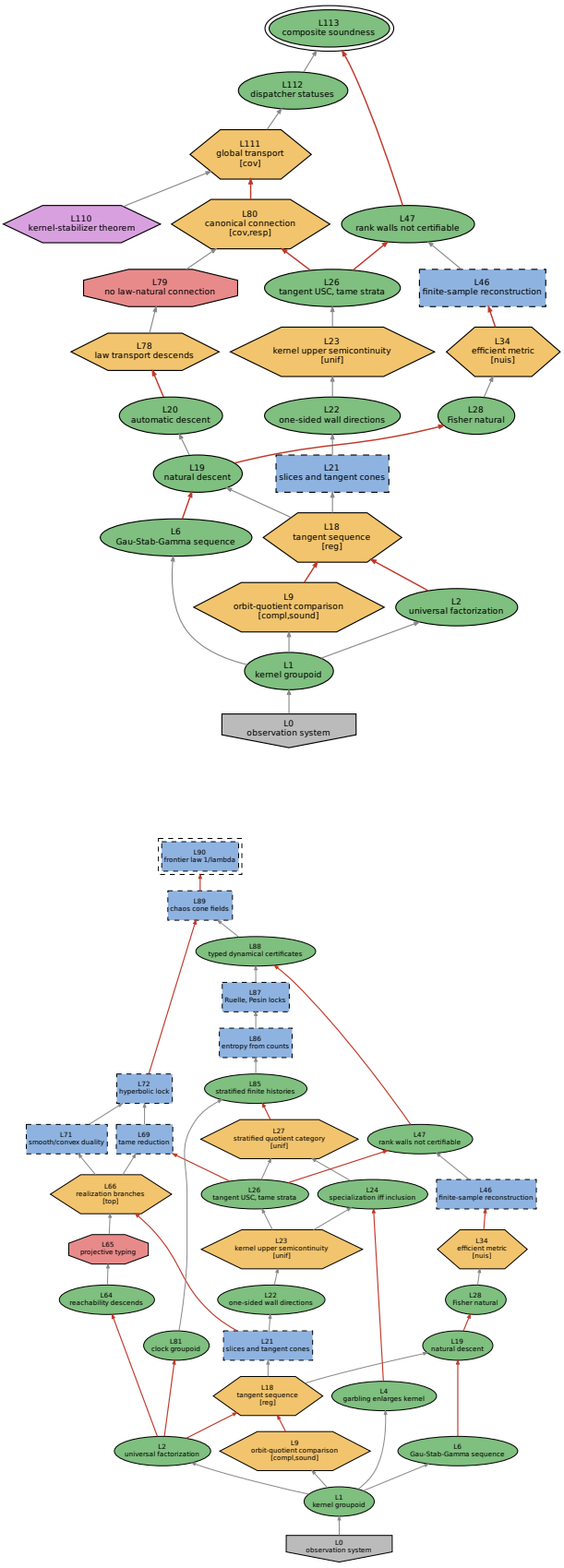


Figure 6: Anti-ladders of L113, composite soundness (laboratory), and L90, the chaos frontier law (physics; the longest chain).

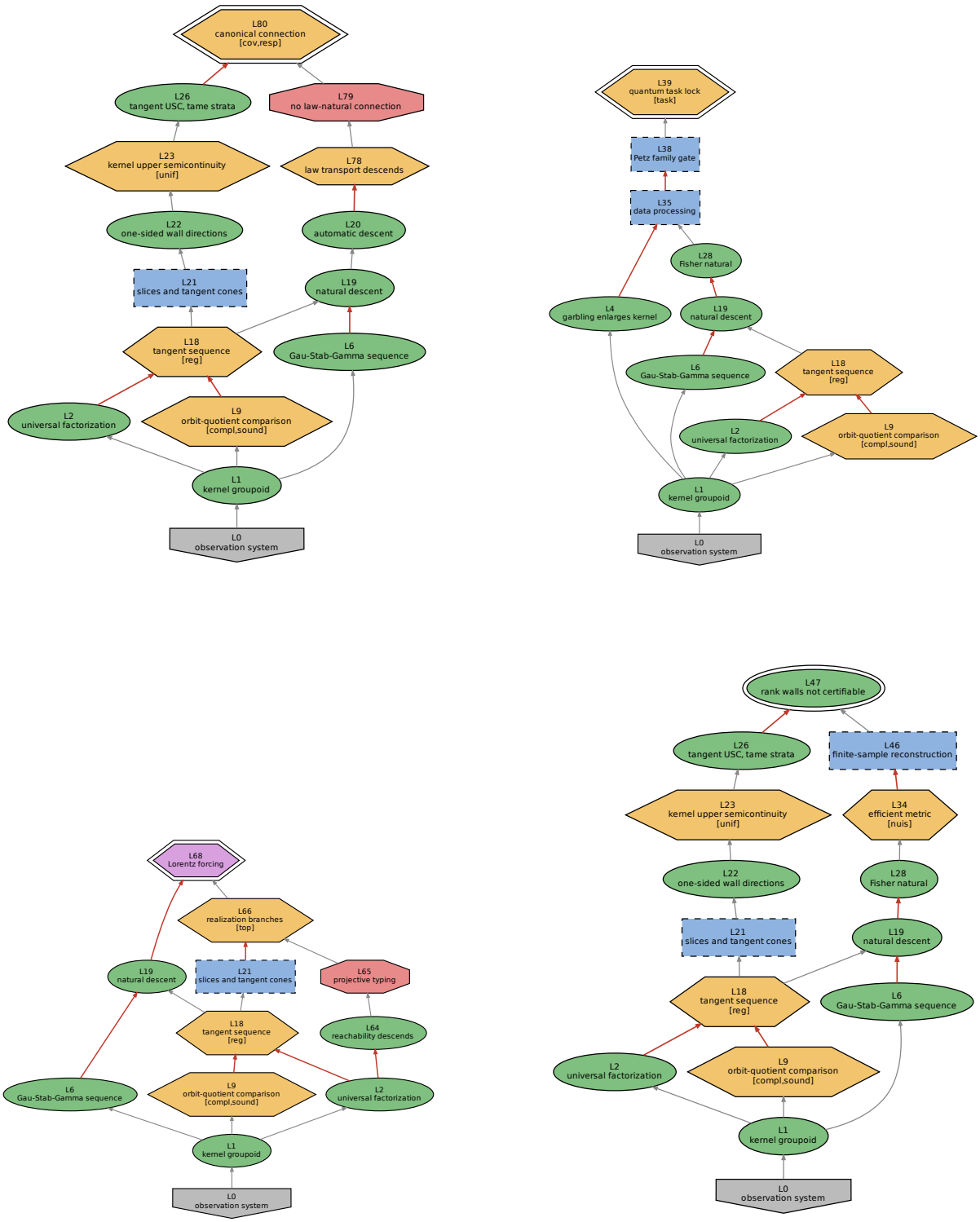


Figure 7: Anti-ladders of L80 (canonical connection), L39 (quantum task lock), L68 (Lorentz forcing) and L47 (rank walls are not certifiable).

9 View 7: audience readings

The same ladder supports three readings. They differ in what is taken as given and in what counts as an output.

9.1 Physics reading: what does the result assume about the world?

Physics enters only through gates: H-SI, H-med, H-val, H-stat, H-resp, H-cov, H-int, Γ (isotropy, Euclidean motions, time flow), and imports such as Landau–Lifshitz and Pesin. The anti-ladders show that every output in the duration, chaos and physical-distance lanes (B10–B12) is reached through hand-overs from the core (B0) and the strata (B2).

Target	Name	Anc.	Chain	Hypothesis budget	Imp.
L109	physical distance	34	11	SI, compl, int, med, nuis, reg, sens, sound, top, val, Γ	5
L105	Sagnac holonomy	19	10	SI, compl, int, reg, sound, top, Γ	3
L68	Lorentz forcing	11	6	compl, reg, sound, top, Γ	1
L80	canonical connection	14	8	compl, cov, reg, resp, sound, unif	2
L90	frontier law 1/lambda	30	14	compl, nuis, reg, sound, top, unif	10
L39	quantum task lock	11	8	compl, reg, sound, task	2
L43	contextual protocols (Fine)	14	9	compl, reg, sound	4
L104	Landau-Lifshitz stationary	18	9	SI, compl, int, reg, sound, top, Γ	3

9.2 Geometry reading: what is forced, what is structure?

The geometer reads the F/I spine, the unconditional core (Figure 4), the strata (B2), the cone (B8) and the remainder (B14). Walls such as L65, L70, L117, L118 and L122 mark where classification stops being unique.

Target	Name	Anc.	Chain	Hypothesis budget	Imp.
L27	stratified quotient category	11	8	compl, reg, sound, unif	2
L32	quotient dim p+q+pq	9	7	compl, reg, sound	1
L69	tame reduction	12	8	compl, reg, sound, top, unif	3
L72	hyperbolic lock	14	9	compl, reg, sound, top, unif	5
L110	kernel-stabilizer theorem	0	0	, Γ	0
L119	global structure	19	10	compl, reg, sound, split, type, unif	4
L121	gluing class H2	20	11	band, compl, reg, sound, split, type, unif	5
L122	wall monodromy	19	10	compl, reg, sound, unif	4

9.3 Laboratory reading: what can be certified, and how does it fail?

The laboratory reads certificates and statuses: L12 (instruments), L16 (confidence sets), L47 (walls), L57 (non-circularity), L113 and L114 (soundness, critical margins). Every failure is typed by the dispatcher (L112), so an anti-ladder doubles as a checklist of the gates whose failure produces each status.

Target	Name	Anc.	Chain	Hypothesis budget	Imp.
L12	minimal-panel selector	6	4	coh, compl, sound	0
L16	confidence-set bridge	9	6	coh, compl, sound	0
L47	rank walls not certifiable	14	8	compl, nuis, reg, sound, unif	3
L57	non-circularity certificate	21	13	adm, compl, nuis, reg, sound, unif	5
L95	counted channel	5	3	—	0
L108	ergodic metre transfer	17	8	SI, compl, med, reg, sens, sound, stat, Γ	0
L113	composite soundness	22	11	compl, cov, nuis, reg, resp, sound, unif, Γ	3
L114	window-indexed walls	23	12	compl, cov, nuis, reg, resp, sound, unif, Γ	3

9.4 Criticality: what falls if a rung fails

Rung	Name	Forward cone	Branches reached
L0	observation system	125	16
L1	kernel groupoid	123	16
L2	universal factorization	108	16
L9	orbit-quotient comparison	102	16
L18	tangent sequence	95	14
L6	Gau-Stab-Gamma sequence	85	16
L19	natural descent	69	14
L21	slices and tangent cones	58	13
L28	Fisher natural	57	11
L22	one-sided wall directions	46	12
L23	kernel upper semicontinuity	45	12
L26	tangent USC, tame strata	42	12
L34	efficient metric	31	9
L4	garbling enlarges kernel	29	9
L46	finite-sample reconstruction	23	7

Table 9: The fifteen rungs with the largest forward cones. L9 ranks fourth, only because of the bundling found in audit A2.

10 View 8: measurement pipelines

A pipeline is a concrete task together with the target rungs it must reach. Its budget is the union of the targets' anti-ladders. The column *walls on the way* lists no-go rungs inside the cone. These are conditions the procedure must respect, and they are where an honest report says CORRESPONDENCE or STRUCTURAL WALL instead of LOCKED.

	Measurement / task	Target rungs	Rungs	Hypothesis budget	Walls on the way
P1	Choose sufficient instruments	L12, L14	8	coh, compl, sound	—
P2	Report a value with uncertainty	L16, L49, L51	22	coh, compl, nuis, reg, sound, unif	—
P3	Certify a wall or a rank	L23, L26, L47, L114	24	compl, cov, nuis, reg, resp, sound, unif, Γ	L79
P4	Calibrated distance (radar, SI)	L102, L103, L107, L109	35	SI, compl, int, med, nuis, reg, sens, sound, top, val, Γ	L65, L94, L96, L106
P5	Compare and synchronize clocks	L82, L83, L93, L105	23	SI, compl, int, reg, sound, top, Γ	L65
P6	Forecast horizon of a chaotic system	L88, L90, L91, L92	33	compl, nuis, reg, sound, top, unif	L65
P7	Non-circular simulation / forgetting	L54, L55, L57	22	adm, compl, nuis, reg, sound, unif	L52, L53, L56
P8	Quantum estimation or testing	L39, L40, L41	21	compl, hell, reg, sound, task, unif	—
P9	Locomotion / latent transport	L80, L111	17	compl, cov, reg, resp, sound, unif, Γ	L79
P10	Lock decision for any request	L112, L113, L114, L115	25	compl, cov, nuis, reg, resp, sound, unif, Γ	L79
P11	Global gluing of local measurements	L119, L121, L122	22	band, compl, reg, sound, split, type, unif	L117, L118, L122

Table 11: Measurement pipelines ("Rungs" is the size of the union cone).

Reading example (P4, calibrated distance). The procedure needs 35 rungs from nine branches. It

assumes H-reg, H-nuis, H-sens, H-top, H-int, SI, med and val, plus Γ, together with H-sound and H-compl through L9 (audit A2). Its cone contains four walls, L65, L94, L96 and L106: no internal unit, count loss, non-stationarity, and projective typing of the cone. A laboratory that cannot certify the medium (H-med) or the held-out anchor (H-val) must report the dilation fibre, not a metre.

11 Companion papers attached to the ladder

Companion	Attaches at	Content
Final Remainder	L2, L15, L27, L30, L32, L115	L116-L126 (incorporated)
Physical Path (Phi0-Phi16)	L28, L30, L32, L80, L117, L119	stress symbol; kinetic cone; defect charge; gravity no-go
Gravity Duality (Psi0-Psi15)	L80, L102, L104, L109	Jacobi pencil (h,N); exact Newton failure point
Light as Emission (Lambda0-Lambda14)	L102, L104, L105, L6	projection formula; mirror quotient; radar round trip
Gravitation Computation	L67, L68, L74, L109	operational metric; declared Einstein closure
Physical Curvature	L80	canonical connection as law of motion (H-resp)

Companion ladders (Φ, Ψ, Λ) are extensions. They attach at the listed rungs and may inherit those rungs' budgets, but they are not part of L0–L126.

12 Audit findings for the next print

The organization exposes three places where a printed dependency carries more conditionality than the result needs. The effect of each proposed rewiring is computed on the graph; none is applied to the print.

ID	Finding and proposal	Rungs affected	Hypotheses removed from their budgets
A1	The dispatcher L112 depends on L111 (global transport), and therefore on L80, which carries H-cov and H-resp. Totality and soundness of the dispatcher should not need latent transport. *Proposal:* L112←L110; L111 becomes a sibling module.	6 (L112–L116, L123)	cov, resp (all six); L112 also sound, compl, reg, unif
A2	L18 (tangent sequence) depends on L9 (orbit–quotient comparison, H-sound, H-compl) only for its completeness-defect clause. *Proposal:* split L18; the tangent sequence needs only L2 and H-reg.	96	sound, compl
A3	L80 carries H-resp only for its physical corollary (Physical Curvature). *Proposal:* move H-resp to a corollary rung, leaving L80 with H-cov.	8	resp

A2 is the largest. As printed, almost the whole differential, second-order, cone, dynamics and physical-distance tower formally inherits the declared-groupoid hypotheses. The proposal changes no theorem; it changes what the ladder says each theorem assumes.

13 Reproducibility

`rg_ladder_org.py` (Python, NetworkX, Matplotlib, Graphviz) reads the ladder table directly from the print source, records its SHA-256, runs checks O1–O9, and regenerates every figure and table in this paper. It also writes `rg_ladder_org.json`, which contains the full register, chains, cones, gates, walls, hand-overs, pipelines and audits. The output is deterministic. Short rung names, branch boundaries,

audience tags, curated wall openers, targets and pipelines are declared in the script; everything else is computed.

Limits. The organization is only as good as the printed dependency column. A missing dependency in the print is invisible here, and an over-stated one appears as inherited conditionality (A1–A3). Transitive reduction changes the drawings, not the analysis. Audience tags are editorial.

A Full register: local versus inherited status

"Anc." is the number of logical ancestors; "Cone" is the forward cone. Rungs downstream of L110 also inherit, through the umbrella, the hypotheses of whichever modules a request uses.

Rung	Name	Br.	Local status	Inherited hypotheses	Anc.	Cone
L0	observation system	B0	input	—	0	125
L1	kernel groupoid	B0	F	—	1	123
L2	universal factorization	B0	F	—	2	108
L3	joint kernel = intersection	B0	F	—	2	19
L4	garbling enlarges kernel	B0	F	—	2	29
L5	exact layer fixes nothing	B0	N	—	3	0
L6	Gau-Stab-Gamma sequence	B0	F	—	2	85
L7	no splitting	B0	N	—	3	0
L8	count rigidity	B0	F	—	1	11
L9	orbit-quotient comparison	B0	H: sound, compl	compl, sound	2	102
L10	rank, separation, completeness distinct	B0	N	compl, sound	4	0
L11	coherent completeness	B0	H: coh	coh, compl, sound	4	4
L12	minimal-panel selector	B1	F	coh, compl, sound	6	3
L13	no task-free selector	B1	N	coh, compl, sound	7	0
L14	robust observability	B1	F	coh, compl, sound	7	1
L15	coupled refines marginals	B1	F	—	3	3
L16	confidence-set bridge	B1	F	coh, compl, sound	9	0
L17	tolerance not a groupoid	B1	N	—	2	0
L18	tangent sequence	B2	H: reg	compl, reg, sound	4	95
L19	natural descent	B2	F	compl, reg, sound	6	69
L20	automatic descent	B2	F	compl, reg, sound	7	10
L21	slices and tangent cones	B2	I	compl, reg, sound	5	58
L22	one-sided wall directions	B2	F	compl, reg, sound	6	46
L23	kernel upper semicontinuity	B2	H: unif	compl, reg, sound, unif	7	45
L24	specialization iff inclusion	B2	F	compl, reg, sound, unif	9	19
L25	normalizer descent	B2	F	compl, reg, sound, unif	11	0
L26	tangent USC, tame strata	B2	F / I	compl, reg, sound, unif	8	42
L27	stratified quotient category	B2	H: unif	compl, reg, sound, unif	11	17
L28	Fisher natural	B3	F	compl, reg, sound	7	57
L29	Fisher selected (Cencov)	B3	N / H: mono	compl, mono, reg, sound	8	0
L30	(R,N,Sigma) reconstruction	B3	I	compl, reg, sound	8	11
L31	Fisher does not fix covariance	B3	N	compl, reg, sound	9	2
L32	quotient dim p+q+pq	B3	F	compl, reg, sound	9	9
L33	shadow is not the law	B3	N	compl, reg, sound	10	0
L34	efficient metric	B3	H: nuis	compl, nuis, reg, sound	8	31
L35	data processing	B3	I	compl, reg, sound	9	11
L36	kernel order not Blackwell	B3	N	compl, reg, sound	10	0
L37	Hellinger specialization	B3	H: unif, hell	compl, hell, reg, sound, unif	16	2
L38	Petz family gate	B4	I	compl, reg, sound	10	5
L39	quantum task lock	B4	H: task	compl, reg, sound, task	11	1
L40	Petz identification	B4	F	compl, reg, sound	11	0
L41	quantum wall f(0)¿0	B4	F	compl, hell, reg, sound, task, unif	19	0
L42	trace rule	B4	I	compl, reg, sound	11	1
L43	contextual protocols (Fine)	B4	I / F	compl, reg, sound	14	0
L44	persistent null	B5	H: sens	compl, reg, sens, sound	9	4
L45	sensor/precision/representation nulls	B5	F	compl, reg, sens, sound	10	0
L46	finite-sample reconstruction	B5	I	compl, nuis, reg, sound	9	23
L47	rank walls not certifiable	B5	F	compl, nuis, reg, sound, unif	14	22
L48	law equality not certifiable	B5	N	compl, nuis, reg, sound, unif	15	0
L49	uncertainty splits on quotient	B6	F / I	compl, nuis, reg, sound, unif	15	10
L50	de Finetti	B6	H: exch	compl, exch, nuis, reg, sound, unif	16	0
L51	LAN and folds	B6	I / F	compl, nuis, reg, sound, unif	16	0

Rung	Name	Br.	Local status	Inherited hypotheses	Anc.	Cone
L52	TV rigidity	B6	N	compl, nuis, reg, sound, unif	16	6
L53	no uniform forgetting	B6	N	compl, nuis, reg, sound, unif	17	5
L54	BV mixing lock	B6	H: adm	adm, compl, nuis, reg, sound, unif	18	1
L55	contraction gate	B6	H: adm	adm, compl, nuis, reg, sound, unif	18	2
L56	samples refute, operators lock	B6	I / N	compl, nuis, reg, sound, unif	18	1
L57	non-circularity certificate	B6	H: adm	adm, compl, nuis, reg, sound, unif	21	0
L58	non-dominated strata	B6	I / F	compl, hell, nuis, reg, sound, unif	21	0
L59	RG distance	B7	H: reg, nuis	compl, nuis, reg, sound	9	6
L60	garbling monotone	B7	F	compl, nuis, reg, sound	12	1
L61	no angle in rank one	B7	N	compl, nuis, reg, sound	10	0
L62	divergences not rulers	B7	N	compl, nuis, reg, sound	10	0
L63	experiment-relative distance	B7	N	compl, nuis, reg, sound	13	0
L64	reachability descends	B8	F	—	3	21
L65	projective typing	B8	N	—	4	20
L66	realization branches	B8	H: top	compl, reg, sound, top	8	19
L67	conformal ray	B8	I	compl, reg, sound, top	9	7
L68	Lorentz forcing	B8	Γ	compl, reg, sound, top, Γ	11	0
L69	tame reduction	B8	I / F	compl, reg, sound, top, unif	12	8
L70	no finite classification	B8	N	compl, reg, sound, top, unif	13	0
L71	smooth/convex duality	B8	I	compl, reg, sound, top	9	5
L72	hyperbolic lock	B8	I / F	compl, reg, sound, top, unif	14	4
L73	wild class	B8	N	compl, reg, sound, top, unif	13	0
L74	scale cocycle	B9	H: int	compl, int, reg, sound, top	10	6
L75	strict transport	B9	F	compl, int, reg, sound, top	11	0
L76	induced connection	B9	I	compl, reg, sound	5	0
L77	entropy selects no connection	B9	N	—	3	0
L78	law transport descends	B9	F / H	compl, reg, sound	8	9
L79	no law-natural connection	B9	N	compl, reg, sound	9	8
L80	canonical connection	B9	H: cov; resp	compl, cov, reg, resp, sound, unif	14	7
L81	clock groupoid	B10	F	—	3	20
L82	affine clocks	B10	Γ	, Γ	5	1
L83	path independence	B10	F	—	4	0
L84	coherence duration (BCH)	B10	I	—	4	0
L85	stratified finite histories	B10	F	compl, reg, sound, unif	13	7
L86	entropy from counts	B10	I	compl, reg, sound, unif	14	6
L87	Ruelle, Pesin locks	B10	I	compl, reg, sound, unif	15	5
L88	typed dynamical certificates	B10	F	compl, nuis, reg, sound, unif	22	4
L89	chaos cone fields	B10	I	compl, nuis, reg, sound, top, unif	29	3
L90	frontier law 1/lambda	B11	I / F	compl, nuis, reg, sound, top, unif	30	2
L91	duality 2 lambda, growth h	B11	F	compl, nuis, reg, sound, top, unif	31	1
L92	lambda/ln2 bits per step	B11	I	compl, nuis, reg, sound, top, unif	32	0
L93	offset/rate/warp	B11	F	—	5	0
L94	no internal unit	B12	N	compl, nuis, reg, sound	10	1
L95	counted channel	B12	F	—	5	6
L96	count loss	B12	N	—	6	1
L97	ratio scale	B12	I	—	3	1
L98	two-point invariant	B12	Γ	compl, reg, sound, Γ	7	5
L99	metric proportionality	B12	Γ	compl, reg, sound, Γ	8	4
L100	anchor budget	B12	Γ	compl, reg, sound, Γ	9	1
L101	unit-distance rigidity	B12	I	compl, reg, sound, Γ	9	1
L102	radar distance	B12	H: SI	SI	5	7
L103	radar = rod	B12	Γ	SI, compl, int, reg, sound, top, Γ	17	4
L104	Landau-Lifshitz stationary	B12	I	SI, compl, int, reg, sound, top, Γ	18	3
L105	Sagnac holonomy	B12	F	SI, compl, int, reg, sound, top, Γ	19	1

Rung	Name	Br.	Local status	Inherited hypotheses	Anc.	Cone
L106	non-stationary no-go	B12	N	SI, compl, int, reg, sound, top, Γ	19	1
L107	media	B12	H: med	SI, compl, med, reg, sens, sound	13	2
L108	ergodic metre transfer	B12	H: stat	SI, compl, med, reg, sens, sound, stat, Γ	17	0
L109	physical distance	B12	H: reg, nuis, SI, med, val; Γ	SI, compl, int, med, nuis, reg, sens, sound, top, val, Γ	34	0
L110	kernel-stabilizer theorem	B13	Γ	, Γ	0	7
L111	global transport	B13	H: cov	compl, cov, reg, resp, sound, unif, Γ	16	6
L112	dispatcher statuses	B13	F	compl, cov, reg, resp, sound, unif, Γ	17	5
L113	composite soundness	B13	F	compl, cov, nuis, reg, resp, sound, unif, Γ	22	4
L114	window-indexed walls	B13	F	compl, cov, nuis, reg, resp, sound, unif, Γ	23	3
L115	maximal closure	B13	F	compl, cov, nuis, reg, resp, sound, unif, Γ	24	2
L116	task residual	B14	F	compl, cov, nuis, reg, resp, sound, unif, Γ	27	1
L117	coupling complement-relative	B14	N	compl, reg, sound	10	7
L118	coupling stabilizer	B14	I / N	compl, reg, sound	11	6
L119	global structure	B14	H: split, type	compl, reg, sound, split, type, unif	19	4
L120	remainder consequences	B14	F	compl, reg, sound, split, type, unif	21	1
L121	gluing class H2	B14	I / H: band	band, compl, reg, sound, split, type, unif	20	1
L122	wall monodromy	B14	N / F	compl, reg, sound, unif	19	2
L123	no new primitive	B14	F (H: split, type)	band, compl, cov, nuis, reg, resp, sound, split, type, unif, Γ	40	0
L124	tame type enumeration	B15	O	compl, reg, sound, top, unif	13	0
L125	forced turbulence forgetting	B15	O	adm, compl, nuis, reg, sound, unif	19	0
L126	remainder across walls	B15	O	compl, reg, sound, split, type, unif	21	0